



IECQ COMPONENT SPECIFICATION

IEC Quality Assessment System for Electronic Components (IECQ System)

Touch panels: Resistive touch panel



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IECQ CS 033200-TW0001

Edition 1.0 2014-06

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INTERNATIONAL
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FOREWORD

The IEC Quality Assessment System for Electronic Components (IECQ) is composed of those member countries of the International Electrotechnical Commission (IEC) who wish to take part in a harmonized system for electronic components of assessment quality. IECQ is also formally known in some European member countries as IECQ-CECC.

The object of the System is to facilitate international trade via business to business supply chain management tools and the harmonization of the specifications and quality assessment procedures for electronic components, assemblies and related materials and processes, and by the grant of an international recognized Certification of Conformity and the optional use of an IECQ Mark of Conformity. The components produced or services provided under the System are therefore accepted in all member countries without further testing.

This Component Specification is based upon the requirements of IECQ 03 Series of Rules of Procedure by:

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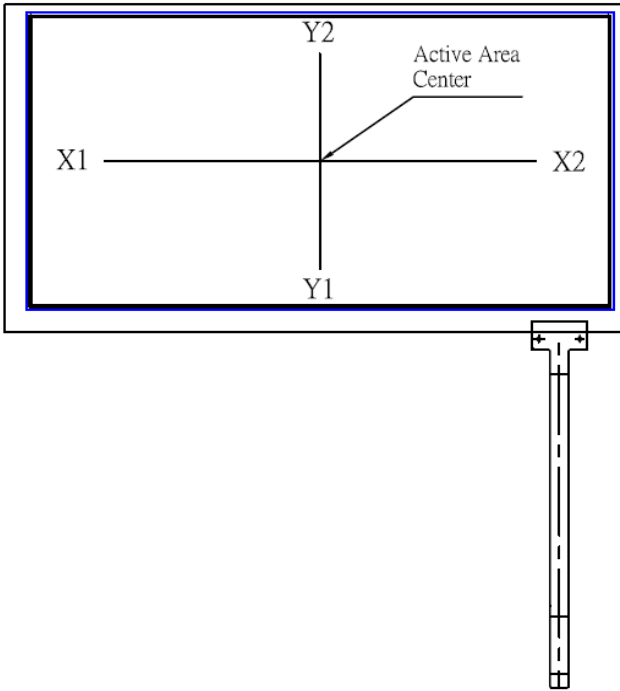
AMENDMENT RECORD

No previous editions.

REQUIREMENTS

The following data sheet satisfies the requirements of IECQ Component Specifications as detailed in IECQ 03 Series of Rules of Procedure.

It should be note that IECQ is not responsible for manufactures declaration made in data sheets which fall outside the limits of Certificates of Conformity.

 Component Specification available from: - UL DQS Taiwan Inc. - IEC Webstore and - IECQ Publications	IECQ CS 033200-TW0001 for use within the IECQ Approved Component Scheme - Automotive Qualification Programme (IECQ AC-AQP)
Electronic Components of Assessed Quality Component Specification in according with: JIS C 5016 8.7, JIS K-5600-5-4 JIS K-5600-5-6, JIS K-7105 JIS Z 0200	Product description Touch Panel: Resistive Touch Panel, Range: Size - 6.5 inch ~ 9 inch Resistance – 10 MΩ ~ 20 MΩ
Outline Drawing and Install Information: 	Applicant: TRANSTOUCH TECHNOLOGY INC. 6F-1, No.50, Hwa-ya 3 Rd. Kwei-san shian, Tao-yuan Taiwan Email: robertteng@transtouch.com.tw Certification Body: DQS-UL MSS (Management Systems Solutions) (Taiwan) Inc.

SECTION ONE – GENERAL INFORMATION

1.1 Approval scope

Table 1

T/P size Electrical characteristics	6.5 inch	6.95 inch	7 inch	8 inch	9 inch
Resistance 10 MΩ			1301-T940 1302-T461 1302-T660 1302-T661		
Resistance 20 MΩ	1301-T690 1302-T410 1302-T420 1302-T610	1303-T080	1301-T660 1301-T900 1302-T180 1302-T182 1302-T183	1301-T922	1302-T253

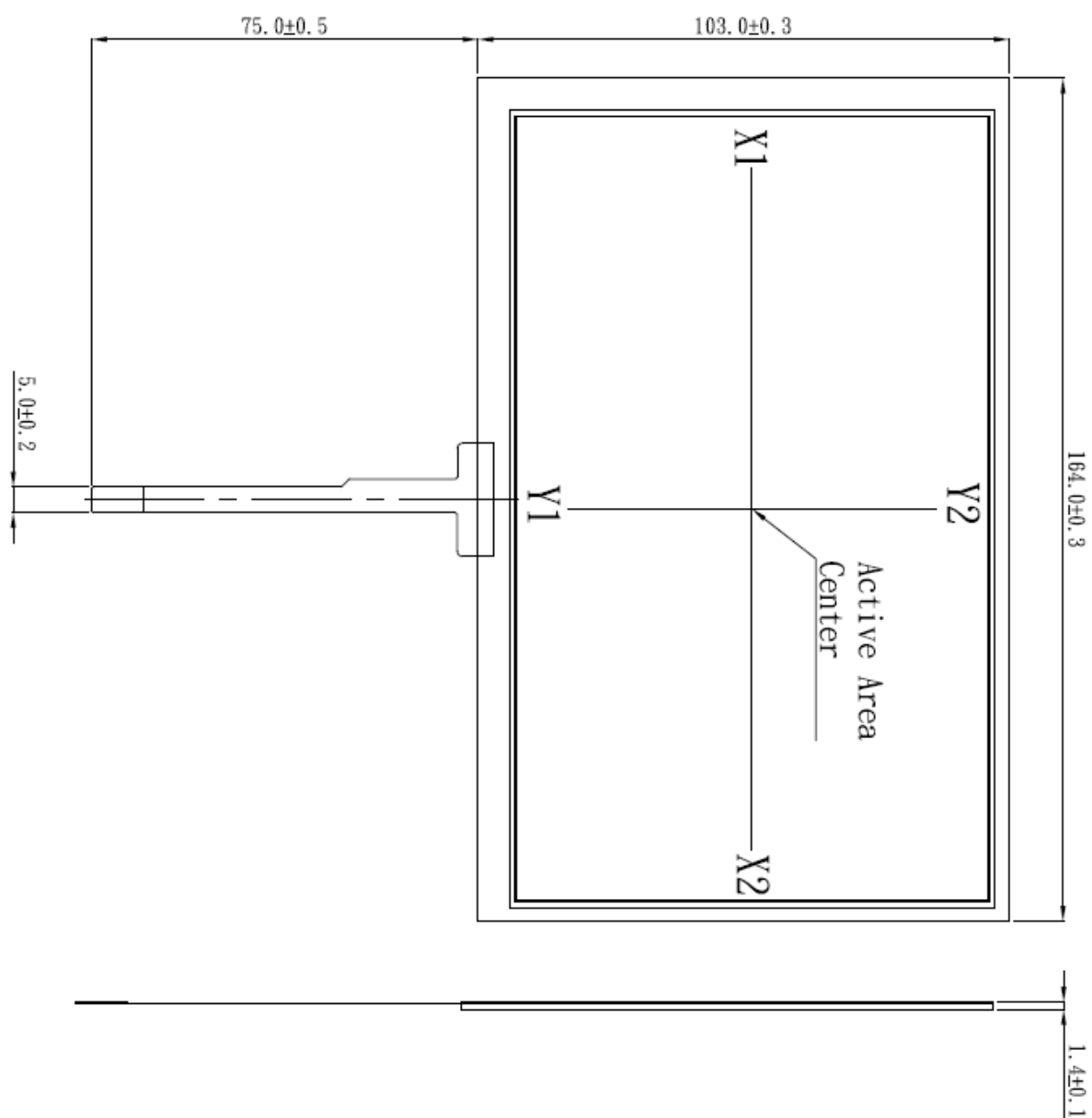
1.2 Outline drawing and dimension**Figure 1**

Table 2

Model no.	Dimension item (unit: mm)				
	X axis	Y axis	Thickness	FPC length	FPC width
T010-1301-T660	164.90±0.3	100.00±0.3	1.80±0.15	50.00±0.5	5.0±0.2
T010-1301-T690	155.00±0.3	89.20±0.3	1.40±0.1	50.00±0.5	5.0±0.2
T010-1301-T900	164.90±0.3	100.00±0.3	1.80±0.15	50.00±0.5	5.0±0.2
T010-1301-T922	192.80±0.3	116.90±0.3	1.40±0.1	70.00±0.5	5.0±0.2
T010-1301-T940	164.00±0.3	103.00±0.3	1.40±0.1	75.00±0.5	5.0±0.2
T010-1302-T180	164.00±0.3	103.00±0.3	1.40±0.1	120.00±0.5	5.0±0.2
T010-1302-T182	164.00±0.3	103.00±0.3	1.40±0.1	120.00±0.5	5.0±0.2
T010-1302-T183	164.00±0.3	103.00±0.3	1.40±0.1	61.00±0.5	5.0±0.2
T010-1302-T253	210.70±0.5	126.40±0.5	1.56±0.15	100.00±0.5	5.0±0.2
T010-1302-T410	155.00±0.3	89.20±0.3	1.40±0.1	35.00±0.5	5.0±0.2
T010-1302-T420	155.00±0.3	89.20±0.3	1.40±0.1	56.50±0.5	5.0±0.2
T010-1302-T461	164.00±0.3	103.00±0.3	1.56±0.15	78.00±0.5	5.0±0.2
T010-1302-T610	155.00±0.3	89.20±0.3	1.40±0.1	50.00±0.5	5.0±0.2
T010-1302-T660	167.00±0.3	94.00±0.3	1.35±0.2	100.00±0.5	5.0±0.2
T010-1302-T661	167.00±0.3	94.00±0.3	1.35±0.2	31.50±0.5	5.0±0.2
T010-1303-T080	166.00±0.3	91.90±0.3	1.40±0.15	100.00±0.5	5.0±0.1

1.3 Limiting value

Table 3

Subclause	Parameter	Symbols	Value		Unit	Test stage
			Min.	Max.		
C2.1, C2.2	Operation temperature	T _{OP}	-35	85	°C	ES
C5.1	Storage temperature	T _{STG}	-40	85	°C	ES
C2.1, C2.2	Operation humidity	Hum _{OP}	10	90	%RH	ES
C5.1	Storage humidity	Hum _{STG}	10	90	%RH	ES
A3.4	Hardness of surface (NOTE 1)	H _S	3		H	IQC
A3.5	Cross-cut test	C _C	0		Level	IQC
A3.1	FPC peeling strength	FPC _{PS}	5		N	FQC
A3.2	FPC bending	FPC _B	3		Times	ES
A3.3	FPC insertion/withdrawal test	FPC _{IW}	5		Times	ES
A2.4	Operation force – Pen	F _P		80	gf	FQC
A2.4	Operation force – Finger	F _F		80	gf	FQC
D1.1	Notes life	N _T	10 ⁵		Words	ES
D1.2	Input life	I _T	10 ⁶		Times	ES

NOTE 1 The test of hardness of surface shall be conducted in incoming inspection of QC department. *Testing Methods for Paints - Part 5: Mechanical Property of Film - Section 4: Scratch Hardness (Pencil Method)*

ES: Engineering sample.

Table 4

Model no.	Optical characteristics				Reference
	Transparency (%)		Haze (%)		
	Standard	Tolerance	Standard	Tolerance	
T010-1301-T660	85	±3	5	±2	JIS-K7105
T010-1301-T690	82	±3	5	±2	
T010-1301-T900	85	±3	5	±2	
T010-1301-T922	80	±3	5	±2	
T010-1301-T940	82	±3	5	±2	
T010-1302-T180	83	±3	5	±2	
T010-1302-T182	83	±3	5	±2	
T010-1302-T183	80	±3	5	±2	
T010-1302-T253	82	±3	5	±2	
T010-1302-T410	82	±3	5	±2	
T010-1302-T420	82	±3	5	±2	
T010-1302-T461	80	±3	6	±2	
T010-1302-T610	82	±3	5	±2	
T010-1302-T660	82	±3	5	±2	
T010-1302-T661	82	±3	5	±2	
T010-1303-T080	80	±3	7	±2	

1.4 Electrical characteristics

Table 5

Test item	Characteristics	Conditions at $T_{op}=26^{\circ}\text{C}\pm 4^{\circ}\text{C}$ unless otherwise stated	Symbols	Requirement		Unit
				Min.	Max.	
	Rated voltage		V_R		5	DC V
B1.1	Resistance	X axis: Film side	R_F	100	1900	Ω
		Y axis: Glass side	R_G	100	1500	
A2.5	Linearity	Initial value	L_{IV}	-1.5	1.5	%
		After RA test	L_{RA}	-3.0	3.0	%
B1.2	Insulation resistance	DC 25V	R_{INS}	20M		Ω
				10M		

RA test: Reliability test

Table 6

Model no.	Electrical characteristics					Related voltage
	Resistance (Ω)		Linearity (max value)		Insulation resistance (Ω) (DC 25V)	
	Film side	Glass side	Initial value	After RA test		
T010-1301-T660	450~1200	100~600	±1.5%	±3.0%	20MΩ min	DC 5V
T010-1301-T690	500~1250	100~550	±1.5%	±3.0%	20MΩ min	
T010-1301-T900	450~1200	100~600	±1.5%	±3.0%	20MΩ min	
T010-1301-T922	150~800	300~1400	±1.5%	±3.0%	20MΩ min	
T010-1301-T940	400~1900	100~900	±1.5%	±3.0%	10MΩ min	
T010-1302-T180	200~1200	400~1500	±1.5%	±3.0%	20MΩ min	
T010-1302-T182	200~1200	400~1500	±1.5%	±3.0%	20MΩ min	
T010-1302-T183	200~1200	400~1500	±1.5%	±3.0%	20MΩ min	
T010-1302-T253	100~800	300~1500	±1.5%	±3.0%	20MΩ min	
T010-1302-T410	500~1250	100~550	±1.5%	±3.0%	20MΩ min	
T010-1302-T420	500~1250	100~550	±1.5%	±3.0%	20MΩ min	
T010-1302-T461	200~1300	100~800	±1.5%	±3.0%	10MΩ min	
T010-1302-T610	500~1900	100~600	±1.5%	±3.0%	20MΩ min	
T010-1302-T660	400~1900	100~650	±1.5%	±3.0%	10MΩ min	
T010-1302-T661	400~1900	100~650	±1.5%	±3.0%	10MΩ min	
T010-1303-T080	450~1500	100~800	±1.5%	±3.0%	20MΩ min	

1.5 Reliability test conditions

Table 7

Model no.	High temperature storage			Low temperature storage			High temperature and high humidity storage				Rapid change of temperature			
	Temperature	Duration	Recovery	Temperature	Duration	Recovery	Temperature	Humidity	Duration	Recovery	Temperature	Cycles	Duration per cycle	Recovery
1301-T660	85°C	240hrs	1hr	-30°C	240hrs	1hr	60°C	95%RH	240hrs	24hrs	-30 ~ 85°C	50 cycles	70mins	24hrs
1301-T690	85°C	240hrs	1hr	-40°C	240hrs	1hr	60°C	95%RH	240hrs	24hrs	-40 ~ 85°C	50 cycles	70mins	24hrs
1301-T900	85°C	240hrs	1hr	-30°C	240hrs	1hr	60°C	95%RH	240hrs	24hrs	-30 ~ 85°C	50 cycles	70mins	24hrs
1301-T922	85°C	240hrs	24hrs	-40°C	240hrs	24hrs	60°C	95%RH	240hrs	24hrs	-40 ~ 85°C	50 cycles	70mins	24hrs
1301-T940	85°C	240hrs	1hr	-40°C	240hrs	1hr	60°C	95%RH	240hrs	24hrs	-40 ~ 85°C	50 cycles	70mins	24hrs
1302-T180	85°C	240hrs	1hr	-40°C	240hrs	1hr	60°C	95%RH	240hrs	24hrs	-40 ~ 85°C	200 cycles	70mins	24hrs
1302-T182	85°C	240hrs	24hrs	-30°C	240hrs	24hrs	60°C	90%RH	240hrs	24hrs	-40 ~ 85°C	50 cycles	70mins	24hrs
1302-T183	85°C	240hrs	24hrs	-30°C	240hrs	24hrs	60°C	90%RH	240hrs	24hrs	-40 ~ 85°C	50 cycles	70mins	24hrs
1302-T253	85°C	240hrs	24hrs	-40°C	240hrs	24hrs	60°C	90%RH	240hrs	24hrs	-40 ~ 85°C	50 cycles	70mins	24hrs
1302-T410	85°C	240hrs	24hrs	-40°C	240hrs	24hrs	60°C	95%RH	240hrs	24hrs	-40 ~ 85°C	50 cycles	70mins	24hrs
1302-T420	85°C	240hrs	1hr	-40°C	240hrs	1hr	60°C	95%RH	240hrs	24hrs	-40 ~ 85°C	50 cycles	70mins	24hrs
1302-T461	85°C	240hrs	24hrs	-40°C	240hrs	24hrs	60°C	95%RH	240hrs	24hrs	-40 ~ 85°C	50 cycles	70mins	24hrs
1302-T610	85°C	240hrs	24hrs	-40°C	240hrs	24hrs	60°C	90%RH	240hrs	24hrs	-40 ~ 85°C	50 cycles	70mins	24hrs
1302-T660	85°C	240hrs	24hrs	-40°C	240hrs	24hrs	60°C	95%RH	240hrs	24hrs	-40 ~ 85°C	100 cycles	70mins	24hrs
1302-T661	85°C	240hrs	24hrs	-40°C	240hrs	24hrs	60°C	95%RH	240hrs	24hrs	-40 ~ 85°C	100 cycles	70mins	24hrs
1303-T080	85°C	240hrs	24hrs	-40°C	240hrs	24hrs	60°C	90%RH	240hrs	24hrs	-40 ~ 85°C	100 cycles	70mins	24hrs

1.6 Related documents

IECQ QC03-1, *Rule of Procedure – Part 1: General Requirements for all IECQ Schemes*

IECQ QC03-3, *Rule of Procedure – Part 3: IECQ Approved Component Products, Related Materials & Assemblies Scheme*

IECQ QC03-3-2: *Rule of Procedure – Part 3-2: IECQ Approved Component Products, Related Materials & Assemblies Scheme, IECQ Approved Component – Automotive Qualification Programme (IECQ AC-AQP)*

AEC-Q200, *Stress Test Qualification for Passive Components*

JIS C 5016 8.7, *Test methods for flexible printed wiring boards*

JIS K-5600-5-4, *Testing Methods for Paints – Part 5: Mechanical Property of Film – Section 4: Scratch Hardness (Pencil Method)*

JIS K-5600-5-6, *Testing Methods for Paints – Part 5: Mechanical Property of Film – Section 6: Adhesion Test (Cross-cut test)*

JIS K-7105, *Japanese Industrial Standard, Testing Methods for Optical Properties of Plastics*

JIS Z 0200, *1999 Packaged freights – General rules of testing*

MIL-STD-1916, *Department of Defense Test Method Standard, DoD Preferred Methods for Acceptance of Product MIL-STD-1916*

The performance tests shall be performed based on the T010-1301-T660 series models' specification.

Table 8

T010-1301-T660 series models' specifications

Model no.	Drawing no.
T010-1301-T660	Tech Bes TTI-06004
T010-1301-T690	Tech Bes TTI-06010
T010-1301-T900	Tech Bes TTI-06062-02
T010-1301-T922	Tech Bes TTI-10074
T010-1301-T940	Tech Bes TTI-07009
T010-1302-T180	Tech Bes TTI-07076-02
T010-1302-T182	Tech Bes TTI-10046-02
T010-1302-T183	Tech Bes TTI-10065
T010-1302-T253	Tech Bes TTI-13010
T010-1302-T410	Tech Bes TTI-08047
T010-1302-T420	Tech Bes TTI-08073
T010-1302-T461	Tech Bes TTI-10030
T010-1302-T610	Tech Bes TTI-08060
T010-1302-T660	Tech Bes TTI-10019
T010-1302-T661	Tech Bes TTI-10051
T010-1303-T080	Tech Bes TTI-10062-N

1.7 Marking

The following mark is to be marked on the outside of package:

- a) Two green RoHS marks stamped on both sides of one full carton
- b) Notification for handle
- c) Packing confirmation stamp
- d) OQC confirmation stamp
- e) Model no. and quantity
- f) Lot no. and carton no.
- g) Other customer's required labels

Model no. and lot no. are printed on the product carton.

1.8 Ordering information

Orders for touch panel covered by this specification shall contain, in clear or in coded form, the following minimum information:

- a) Vendor information
- b) Buyer information
- c) P/O number
- d) Payment term
- e) Delivery term
- f) Model no. or description
- g) Quantity
- h) Unit price and amount
- i) Currency

1.9 Primary stage of manufacture

Primary stage of manufacture in touch panel is lamination of glass and film and Ag pattern print, related parameter and conditions please refer to "TTIW0770509 Operation Conditions for Lamination of Glass and Film" and "TTIW0770503 Operation Conditions for Glass Print". Important characteristic is light input force and without Newton ring.

1.10 Additional information

- a) Since touch panel consists of glass, please be careful your hand and other part from injury at handling. You must wear gloves at handling
- b) Do not put a heavy shock or stress on touch panel
- c) Do not lift touch panel by cable (FPC)
- d) Do not add any stress on film side. (Ex. don't transfer the panel by film side with vacuum)
- e) Please use dry cloth or soft cloth with neutral detergent (after wring dry) or one with ethanol at cleaning. Do not use any organic solvent, acid or alkali solution
- f) Do not pile touch panel. Do not put heavy goods on touch panel
- g) Do not bend a cable of touch panel for prevent happen to line cut failure. Please do not use following method for insert the cable to connector and pay attention to the matter as stated below at mounting design of touch panel and enclosure:
 - 1. Enclosure support to fix touch panel must be out of view (transparent) area. (Do not design enclosure presses the view area to protect from miss input)

2. Enclosure edge must be between view area and guaranteed active area. (Enclosure edge must not touch with view area)
 3. We recommend the material of support to fix touch panel is elastic material
 4. Do not bond top surface (film) of touch panel with enclosure
 5. The corner parts have conductivity. Do not touch any metal part after mounting
 6. Special design is required for water resistance use elastic material
 7. To clean touch panel by air gun, pressure 2 kg/cm² below is suggested. This is preventing FPC to peel off when air is blowing to FPC from glass side
 8. The mounting structure must have a reserved space for the FPC tail and never touch or squeeze the FPC by case or another components preventing FPC to peel off
- h) Inventory which storage in warehouse more than 6 months, the visual inspection and electrical function inspection must be conducted before outbound to customers.

SECTION TWO – QUALITY CONFORMANCE TEST SCHEDULE

2.1 Procedure

2.1.1 For product approval, the procedure shall be in accordance with the Approval Process Plan/Test Plan. Document number: TTIW0670101-1.

2.1.2 For quality conformance inspection, the procedure shall be in accordance with the Detail Specification QC 03-3-2/TW0001.

2.2 Inspection lot and sampling plan

Table 9

Model no.	Inspection information					
	Separate	Lot quantity	Standard	Reduced	Normal	Tightened
T010-1301-T660	4	360pcs	MIL-STD-1916 Level II	6	16	40
T010-1301-T690	6	540pcs		6	16	40
T010-1301-T900	4	360pcs		6	16	40
T010-1301-T922	3	270pcs		5	12	32
T010-1301-T940	4	360pcs		6	16	40
T010-1302-T180	4	360pcs		6	16	40
T010-1302-T182	4	360pcs		6	16	40
T010-1302-T183	4	360pcs		6	16	40
T010-1302-T253	3	270pcs		5	12	32
T010-1302-T410	6	540pcs		6	16	40
T010-1302-T420	6	540pcs		6	16	40
T010-1302-T461	4	360pcs		6	16	40
T010-1302-T610	6	540pcs		6	16	40
T010-1302-T660	6	540pcs		6	16	40
T010-1302-T661	6	540pcs		6	16	40
T010-1303-T080	6	540pcs		6	16	40

2.3 Test samples

2.3.1 In accordance with QC 03-3 Rules of Procedure, Subclause 8.4, Approved Component Testing, the product approval shall be conducted in compliance with this requirement.

2.3.2 For product approval, a fix sample which be defined in the test plan shall be taken from current production of three consecutive lots in as short as possible a period of time.

2.3.3 The inspection level of quality conformance inspection shall be decided accordance with MIL-STD-1916.

2.3.4 The samples which are subjected to conduct the periodic testing have to pass the lot-by-lot tests.

2.3.5 The test methods shall be accordance with international standard, test plan or Section three of this specification.

2.3.6 The measurement shall be carried out after specified time mentioned in Table 7. for recovery of completing the periodic test.

2.4 Quality conformance inspection procedure

2.4.1 The certificate of product approval is maintained after successful completion of all the tests of Detail Specification and/or company's reliability test program.

2.4.2 According to the QC 03-3-2, Subclause 7.3, the items of Group A and Group B are lot-by-lot inspection. Group C and Group D are subjected to periodic tests.

NOTE 1 Subclause number of tests and performance requirements refer to Section three of this specification.

NOTE 2 Inspection levels are selected from MIL-STD-1916.

NOTE 3 The acronym of the Table:

P = periodicity (in months)

n = sample size

c = acceptance criterion (permitted number of defectives)

D = destructive

ND = non destructive

IL = inspection level

Subclause number and test (NOTE 1)	D or ND	Conditions of test (NOTE 1)	IL (NOTE 2)	Performance requirements (NOTE 1)
<u>Group A Inspection</u> (lot-by-lot)				
<u>Sub-group A1</u> A1.1 Visual examination	ND	Test Environment: Temperature =26±4 °C Humidity =40±30%RH Illumination= 700~1000Lux	II	No Scratch and Dent on Glass, Film and Materials between F/G No Scratch and Crumple on FPC.
A1.2 Marking		Test Environment: Temperature =26±4 °C Humidity =40±30%RH Illumination= 700~1000Lux		Legible marking as specified in 1.7
<u>Sub-group A2</u> A2.1 Dimensions	ND	Dimensions shall be measured by caliper	5 pcs	X axis length Y axis length T/P thickness FPC length FPC width as specified in Table 2.
A2.2 Transparency	ND	JIS K-7105 Illumination=higher than 3000Lux	1 pcs	As specified in Section one, Clause 1.3, Table 4.
A2.3 Haze	ND	JIS K-7105 Illumination=higher than 3000Lux	1 pcs	As specified in Section one, Clause 1.3, Table 4.
A2.4 Operation force (g) Finger/Pen	ND	Finger tester=R8 Pen tester=R0.8	5 pcs	As specified in Section one, Clause 1.3, Table 3.
A2.5 Linearity	ND	Test Method Test machine model: FC4200-50 LP-200 XF-500	5 pcs	As specified in Section one, Clause 1.4, Table 6.

Subclause number and test (NOTE 1)	D or ND	Conditions of test (NOTE 1)	IL (NOTE 2)	Performance requirements (NOTE 1)
<u>Sub-group A3</u>				
A3.1 Peeling strength	D		1 pce	≥ 5 N
A3.2 Bending	D	*1 JIS C 5016 8.7	1 pce	3 Times There is no circuit damage and FPC conduction not affected.
A3.3 Insertion/Withdrawal test	D	*1	1 pce	5 Times Acceptable criteria are without scratches and crumple on surface of FPC figure, FPC conduction is not affected.
A3.4 Hardness of surface	D	JIS K-5600-5-4	1 pce	$\geq 3H$
A3.5 Cross-cut test	D	JIS K-5400	1 pce	Test Result = Level 0
<u>Group B Inspection</u> (lot-by-lot)				
<u>Sub-group B1</u>				
Impedance (Ω)				
B1.1 Resistance	ND	Rated Voltage: DC 5V	5 pcs	As specified in Section one, Clause 1.4, Table 6.
B1.2 Insulation resistance		Voltage: DC 25V Duration: 1min	5 pcs	As specified in Section one, Clause 1.4, Table 6.

Subclause number and test (NOTE 1)	D or ND	Conditions of test (NOTE 1)	Sample size & criterion of acceptability (NOTE 3)			Performance requirements (NOTE 1)
			p	n	c	
<u>Group C Inspection</u> (Periodic) <u>Sub-group C1</u> Initial measurements C1.1 Drop test Final measurements	D	Visual examination				No crack, broken, dent on T/P
		Hit points are 1 corner, 3 edges, 6 sides and height of free fall are chosen from Test Method C1.1	*1	*2	0	As specified in Section three Test Method, C1.1
		Visual examination				No crack, broken, dent on T/P
<u>Sub-group C2</u> Initial measurements C2.1 Dynamic test at low temperature and low humidity Final measurements	D	Visual examination Impedance: Resistance Insulation Resistance Force Linearity	*1	10	0	No waving, air leaking and deformation on film side. Other test items F_P , F_F , R_F , R_G , L_{IV} , L_{RA} , and R_{INS} are specified in Table 3 and Table 5.
		Rated Voltage: DC 5V Temperature: -35 °C Humidity: less than 10%RH Duration:240hrs Recovery: 24hrs	*1	5	0	As specified in Table 3.
		Visual examination Impedance: Resistance Insulation Resistance Force Linearity				No waving, air leaking and deformation on film side. Other test items F_P , F_F , R_F , R_G , L_{IV} , L_{RA} , and R_{INS} are specified in Table 3 and Table 5.

Subclause number and test (NOTE 1)	D or ND	Conditions of test (NOTE 1)	Sample size & criterion of acceptability (NOTE 3)			Performance requirements (NOTE 1)
			p	n	c	
C2.2 Dynamic test at high temperature		Rated Voltage: DC 5V Temperature: 85°C Humidity: Less than 10%RH Duration:240hrs Recovery: 24hrs	1*	5	0	
Final measurements		Visual examination Impedance: Resistance Insulation Resistance Force Linearity				No waving, air leaking and deformation on film side. Other test items F_P , F_F , R_F , R_G , L_{IV} , L_{RA} , and R_{INS} are specified in Table 3 and Table 5.
C2.3 Dynamic test at high humidity	D	Rated Voltage: DC 5V Temperature: 25°C±4°C Humidity:90%RH Duration:240hrs Recovery: 24hrs	*1	5	0	
Final measurements		Visual examination Impedance: Resistance Insulation Resistance Force Linearity				No waving, air leaking and deformation on film side. Other test items F_P , F_F , R_F , R_G , L_{IV} , L_{RA} , and R_{INS} are specified in Table 3 and Table 5.

Subclause number and test (NOTE 1)	D or ND	Conditions of test (NOTE 1)	Sample size & criterion of acceptability (NOTE 3)			Performance requirements (NOTE 1)
			p	n	c	
Sub-group C3 Initial measurements	D	Visual examination Impedance: Resistance Insulation Resistance Force Linearity	12	5	0	No waving, air leaking and deformation on film side. Other test items F_P , F_F , R_F , R_G , L_{IV} , L_{RA} , and R_{INS} are specified in Table 3 and Table 5.
C3.1 Rapid change of temperature		Lower Category Temperature: -40°C / -30°C Upper Category Temperature: 85°C 200 cycles, 60min/per cycle *3 Transition time : less than 5 mins. Recovery: 24hrs				As specified in Table 7.
Final measurements		Visual examination Impedance: Resistance Insulation Resistance Force Linearity				No waving, air leaking and deformation on film side. Other test items F_P , F_F , R_F , R_G , L_{IV} , L_{RA} , and R_{INS} are specified in Table 3 and Table 5.

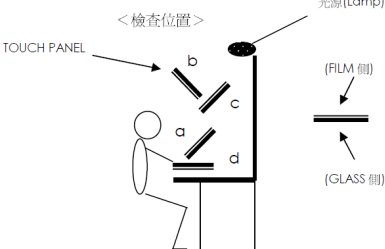
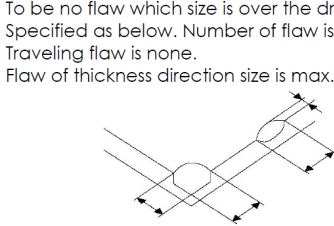
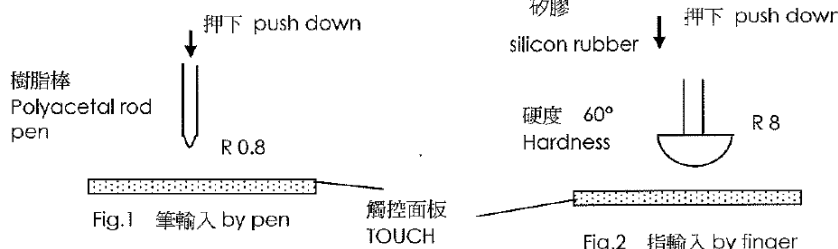
Subclause number and test (NOTE 1)	D or ND	Conditions of test (NOTE 1)	Sample size & criterion of acceptability (NOTE 3)			Performance requirements (NOTE 1)
			p	n	c	
Sub-group C4 Initial measurements	D D	Visual examination Impedance: Resistance Insulation Resistance Force Linearity	12	10	0	No waving, air leaking and deformation on film side. Other test items F_P , F_F , R_F , R_G , L_{IV} , L_{RA} , and R_{INS} are specified in Table 3 and Table 5.
C4.1 Storage at low temperature and low humidity		Temperature: -40°C / -30°C *4 Humidity: less than 10%RH Duration: 240hrs Recovery: 24hrs / 1hr	12	5	0	As specified in Table 7.
Final measurements		Visual examination Impedance: Resistance Insulation Resistance Force Linearity				No waving, air leaking and deformation on film side. Other test items F_P , F_F , R_F , R_G , L_{IV} , L_{RA} , and R_{INS} are specified in Table 3 and Table 5.
C4.2 Storage at high temperature	D	Higher Temperature: 85°C Duration: 240hrs Recovery: 24hrs / 1hr	12	5	0	As specified in Table 7.
Final measurements		Visual examination Impedance: Resistance Insulation Resistance Force Linearity				No waving, air leaking and deformation on film side. Other test items F_P , F_F , R_F , R_G , L_{IV} , L_{RA} , and R_{INS} are specified in Table 3 and Table 5.

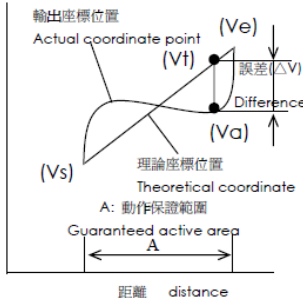
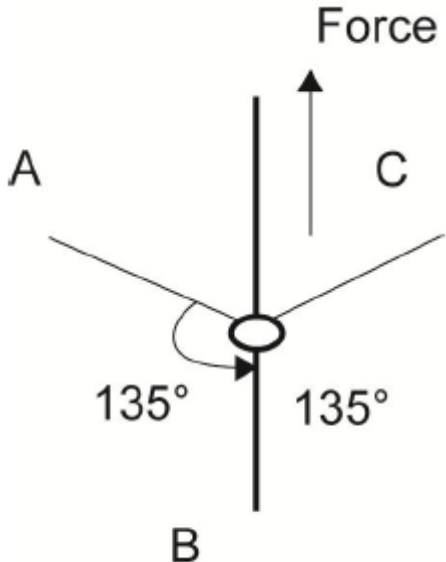
Subclause number and test (NOTE 1)	D or ND	Conditions of test (NOTE 1)	Sample size & criterion of acceptability (NOTE 3)			Performance requirements (NOTE 1)
			p	n	c	
Sub-group C5 Initial measurements	D	Visual examination Impedance: Resistance Insulation Resistance Force Linearity	12	5	0	No waving, air leaking and deformation on film side. Other test items F_P , F_F , R_F , R_G , L_{IV} , L_{RA} , and R_{INS} are specified in Table 3 and Table 5.
C5.1 Storage at high temperature and high humidity		Temperature: 60°C Humidity: 95%RH / 90%RH*5 Duration: 240hrs Recovery: 24hrs	12	5	0	As specified in Table 7.
Final measurements		Visual examination Impedance: Resistance Insulation Resistance Force Linearity				No waving, air leaking and deformation on film side. Other test items F_P , F_F , R_F , R_G , L_{IV} , L_{RA} , and R_{INS} are specified in Table 3 and Table 5.
Group D Inspection (Periodic)	D		*1	2	0	
Sub-group D1 Life test						
D1.1 Input life		$\geq 10^6$ times Input Frequency : 5 times/sec		1	0	F_P , F_F , R_{INS} , R_F , R_G must satisfy specification in Table 3 and Table 5.
D1.2 Notes life		$\geq 10^5$ words Note Speed: 60 mm/sec		1	0	F_P , F_F , R_{INS} , R_F , R_G must satisfy specification in Table 3 and Table 5.

- *1 The test shall be conducted during product approval.
- *2 Quantity of drop test is one carton.
- *3: The number of rapid change is from 50 to 200 cycles, 70 mins/per cycle as specified in Table 7.
- *4: -40°C is the lowest storage temperature, other test temperature as specified in Table 7.
- *5: $60^{\circ}\text{C}/95\%\text{RH}$ is the normal test conditions, other test conditions as specified in Table 7.

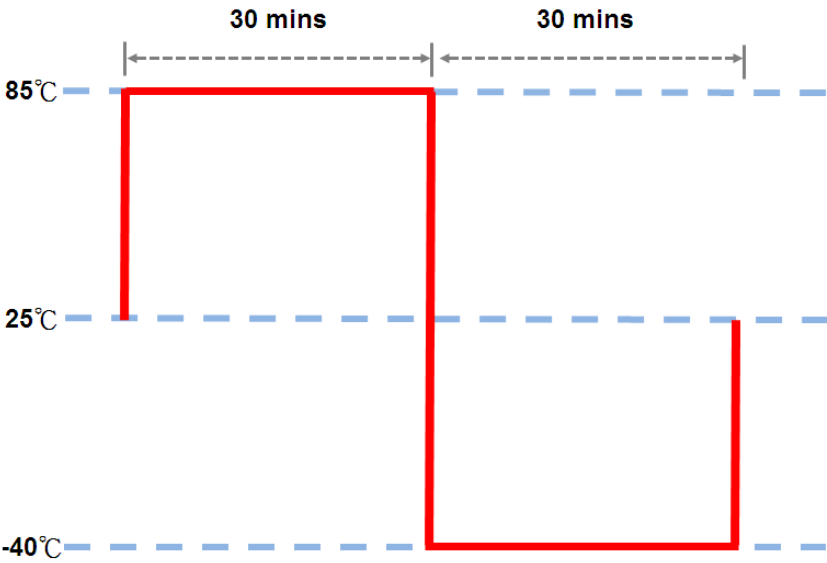
SECTION THREE – TEST METHOD

3.1 Table of test methods for touch panel

Test item	Stress	Reference	Additional requirement
A1.1	Visual examination	Sampling Standard MIL-STD-1916	Film/Glass scratch, Materials between film and glass, Film fish eye, Newton ring, Glass flaw, Film size out spec, FPC material, FPC scratch, FPC crumple, DSA size out spec.
		Visual Exam. Point 	Glass Flaw Judgments To be no flaw which size is over the drawing Specified as below. Number of flaw is none-specify. Traveling flaw is none. Flaw of thickness direction size is max. 
A1.2	Marking	T010-1301-T66 0 Series Models' Specifications and Laser Marking Specification	Laser marking operating condition follow the TTIW0770531 Laser Marking Specification. Laser marking must clear and easy to identify.
A2.1	Dimensions	T010-1301-T66 0 Series Models' Specifications	Use caliper to measure dimensions as specified in Section one, Clause 1.2, Figure 1 and Table 2.
A2.2	Transparency	T010-1301-T66 0 Series Models' Specifications and JIS K-7105	Use Transparency Test Machine to inspect T/P and follow value which specified in Section one, Clause 1.3, Table 4. Follow specification of JIS K-7105. Test Illumination>3000LUX Before testing, make sure the initial value is completed re-calibration.
A2.3	Haze	T010-1301-T66 0 Series Models' Specifications and JIS K-7105	Use Transparency Test Machine to inspect T/P and follow value which specified in Section one, Clause 1.3, Table 4. Follow specification of JIS K-7105. Test Illumination>3000LUX Before testing, make sure the initial value is completed re-calibration.
A2.4	Operation force (Finger/Pen)	As specified in Table 3.	Use Force Gauge to test T/P and follow value which specified in Section one, Clause 1.3, Table 3. The Max. force is 80gf.
			

Test item	Stress	Reference	Additional requirement
A2.5	Linearity	As specified in Table 6.	Use Linearity Test Machine to test T/P and follow value which specified in Section one, Clause 1.4, Table 6. Test Machine Models:FC4200-50, LP-200, XF-500
		直線性測定條件 Measurement condition of Linearity 在動作保證範圍內 2mm，沿著 X、Y 各軸的直線上任意複數點將最外端的點連結成一假想直線，相對於此線的偏差值，直線性是基準電壓分割值得最大偏差電壓。 Difference between actual voltage & Theoretical voltage is an error at any points. Linearity is the value max. error voltage divided by voltage difference on active area inside 2mm. 直線性 Linearity = $\text{Max}(\Delta V) / (V_e - V_s)$ 誤差 Difference $\Delta V = V_a - V_t$ Vs：起始電壓 Start point voltage Ve：終點電壓 End point voltage Va：實際電壓 Actual voltage Vt：理論電壓 Theoretical voltage The Difference between actual voltage and theoretical voltage is calculated at any points. Linearity is the value of max difference voltage on active area.	
A3.1	Peeling strength	As specified in Table 3.	Use Force Measurement Gauges to inspect FPC bonding stable. The peeling angle is upward by 90°.
A3.2	Bending	JIS C 5016 8.7	1. Follow the cycle in Figure 2, from A-B-C-C-B-A. 2. Bending Radius: 1.0mm 3. Angel: 90°±5° 4. Weight: 500g 5. Bending Frequency: 170 times/min 6. There is no circuit damage and FPC conduction not affected.
			Figure 2 

Test item	Stress	Reference	Additional requirement																																							
A3.3	Insertion and withdrawal test	As specified in Table 3.	Use connectors that provided by customers to conduct Insertion and Withdrawal Test 5 times. Acceptable criteria are without scratches and crumple on surface of FPC figure, FPC conduction is not affected.																																							
B1.1	Resistance	As specified in Table 6.	Use Multimeter to measure glass side resistance and film side resistance. Rated Voltage=DC 5V.																																							
B1.2	Insulation resistance	As specified in Table 6.	Use High Resistance Meter to measure insulation resistance. Rated Voltage=DC 25V. Test Duration=1 min.																																							
C1.1	Drop test	JIS C0041-1995	Follow the Specification of JIS C0041-1995 to conduct drop test. Hit points are 1 corner, 3 edges, 6 sides and height of free fall are chosen from list below.																																							
		表 5 落下高さ（自由落下）																																								
		<table><tr><th rowspan="2">総質量 kg</th><th colspan="4">落下高さ cm</th></tr><tr><th>レベル I</th><th>レベル II</th><th>レベル III</th><th>レベル IV</th></tr><tr><td>10 未満</td><td>80</td><td>60</td><td>40</td><td>30</td></tr><tr><td>10 以上 20 未満</td><td>60</td><td>55</td><td>35</td><td>25</td></tr><tr><td>20 以上 30 未満</td><td>50</td><td>45</td><td>30</td><td>20</td></tr><tr><td>30 以上 40 未満</td><td>40</td><td>35</td><td>25</td><td>15</td></tr><tr><td>40 以上 50 未満</td><td>30</td><td>25</td><td>20</td><td>10</td></tr><tr><td>50 以上 100 未満</td><td>25</td><td>20</td><td>15</td><td>10</td></tr></table>		総質量 kg	落下高さ cm				レベル I	レベル II	レベル III	レベル IV	10 未満	80	60	40	30	10 以上 20 未満	60	55	35	25	20 以上 30 未満	50	45	30	20	30 以上 40 未満	40	35	25	15	40 以上 50 未満	30	25	20	10	50 以上 100 未満	25	20	15	10
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落下順序及び落下回数は、表 6 のとおりとする。																																										
表 6 直方体の落下順序及び落下回数																																										
<table><tr><th>落下の順序</th><th>落下の箇所</th><th>回数</th></tr><tr><td>1</td><td>下面に接する角（かど） 例 2-3-5 角</td><td>1</td></tr><tr><td>2</td><td>下面とつま面とに接するりょう 例 3-5 りょう</td><td>1</td></tr><tr><td>3</td><td>下面と側面とに接するりょう 例 2-3 りょう</td><td>1</td></tr><tr><td>4</td><td>側面とつま面とに接するりょう 例 2-5 りょう</td><td>1</td></tr><tr><td>5~10</td><td>6面のすべて</td><td>6</td></tr><tr><td></td><td>計</td><td>10</td></tr></table>		落下の順序	落下の箇所	回数	1	下面に接する角（かど） 例 2-3-5 角	1	2	下面とつま面とに接するりょう 例 3-5 りょう	1	3	下面と側面とに接するりょう 例 2-3 りょう	1	4	側面とつま面とに接するりょう 例 2-5 りょう	1	5~10	6面のすべて	6		計	10																				
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	計	10																																								
C2.1	Dynamic test at low temperature and low humidity	T010-1301-T66 0 Series Models' Specifications	Use thermal shock tester to conduct this RA test. Rated Voltage=DC 5V Temperature=-30℃ Humidity=less than 10%RH Duration=240hrs/No temporary test Test performance after 24hrs recovery.																																							
C2.2	Dynamic test at high temperature	T010-1301-T66 0 Series Models' Specifications	Use thermal shock tester to conduct this RA test. Rated Voltage=DC 5V Temperature=85℃ Humidity=less than 10%RH Duration=240hrs/No temporary test Test performance after 24hrs recovery.																																							

Test item	Stress	Reference	Additional requirement
C2.3	Dynamic test at high humidity	T010-1301-T66 0 Series Models' Specifications	Use thermal shock tester to conduct this RA test. Rated Voltage=DC 5V Temperature=25°C±4°C Humidity=90%RH Duration=240hrs/No temporary test Test performance after 24hrs recovery.
C3.1	Rapid change of temperature	Section one, Clause 1.5, Table 7.	Use thermal shock tester to conduct this RA test. Temperature range is -40°C ~ 85°C, duration 60mins each cycle. Cycles as specified in Table 7. Test performance after 24hrs recovery.
		 The diagram illustrates a thermal shock test cycle. It features three horizontal dashed blue lines representing temperature levels: 85°C at the top, 25°C in the middle, and -40°C at the bottom. A solid red line shows the temperature profile. It starts at 25°C, rises vertically to 85°C, and is held there for 30 minutes (indicated by a horizontal double-headed arrow). It then drops vertically to -40°C and is held there for 30 minutes (also indicated by a horizontal double-headed arrow). Finally, it drops vertically back to 25°C. The cycle is labeled '30 mins' for both the high and low temperature dwell periods.	
C4.1	Storage at low temperature and low humidity	T010-1301-T66 0 Series Models' Specifications	Use thermal shock tester to conduct this RA test. Temperature, Humidity, Duration, Recovery as specified in Table 7.
C4.2	Storage at high temperature	T010-1301-T66 0 Series Models' Specifications	Use thermal shock tester to conduct this RA test. Temperature, Humidity, Duration, Recovery as specified in Table 7.
C5.1	Storage at high temperature and high humidity	T010-1301-T66 0 Series Models' Specifications	Use thermal shock tester to conduct this RA test. Temperature, Humidity, Duration, Recovery as specified in Table 7.
D1.1	Input life test	T010-1301-T66 0 Series Models' Specifications	Use input test machine to test the centre of guaranteed active area. By silicone rubber tapping at same point. Sharp of rubber end=R8 Hardness=60 Load=200g Frequency=5 times/sec ≥ 10 ⁶ times
D1.2	Notes life test	T010-1301-T66 0 Series Models' Specifications	Use Note test machine to test the centre of guaranteed active area. To draw the diagonal back and forth with the speed of 60mm/sec, load 250g. ≥ 10 ⁵ times

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